LING406_Lec14

Course: LING 406: Psycholinguistics

Title of Lecture: Language Development 1: Speech Perception

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Section 1: Nativist Positions in Child Language

- The question of **nature and nurture in child language** asks how much of language comes built into the child and how much is learned from experience. The **"nativist" positions** grew out of **Chomsky's critique of Skinner**, where Chomsky argued that the behaviorist account of learning could not explain how children acquire language.
    - **Skinner** was a behaviorist, and behaviorists studied general learning mechanisms such as reward and punishment. Chomsky's critique was aimed at the idea that these general mechanisms are enough to produce language.
- For Chomsky, the most important feature of language is its **generativity**. Generativity means that a language can produce an unlimited number of new sentences, including sentences that no one has ever said or heard before.
    - From this point of view, **acquiring a language** means identifying the **abstract grammar** that produces the sentences of that language. The child is not just memorizing the sentences heard so far. The child has to work out the hidden system of rules that could generate all of the possible sentences.
- According to the nativist argument, the **general learning mechanisms** studied by behaviorists were not up to this task, and there are two reasons for that claim.
    - The first reason is that **grammatical and ungrammatical sentences are not "similar" in any straightforward way**. A learning mechanism that groups things by how much they resemble each other cannot easily separate the good sentences from the bad ones, since a grammatical sentence and an ungrammatical one can look and sound very much alike.
    - The second reason is that the context for language acquisition does not look like **operant conditioning**. In operant conditioning, a behavior becomes more or less frequent based on whether it is followed by a reward or a punishment, and in language acquisition these **reward and punishment contingencies don't hold.** Children are not systematically rewarded for grammatical sentences and punished for ungrammatical ones, so this kind of learning cannot be what drives grammar acquisition.
- Instead of general learning, nativist approaches assert two things that the child brings to the task.
    - The first is **innate knowledge of what language is like**, which is called **Universal Grammar**. This is knowledge about the general shape of human languages that the child has before any experience with a particular language.
    - The second is **language-specific acquisition processes**. These are learning processes that exist specifically for language rather than being the same general mechanisms used for learning everything else.
- The nativist position is captured in the following quotation: "The fact that all normal children acquire essentially comparable grammars of great complexity with remarkable rapidity suggests that human beings are somehow specially designed to do this, with data-handling or 'hypothesis-formulating' ability of unknown character and complexity."
    - The argument in this quotation rests on three observations. All normal children end up with essentially comparable grammars, those grammars are of great

complexity, and children acquire them with remarkable rapidity.
    - From those observations, the conclusion is that humans are **specially designed** for language acquisition. The quotation describes this special design as a **"data-handling" or "hypothesis-formulating" ability**, meaning a built-in capacity for processing language input and forming guesses about its grammar, while admitting that the exact character and complexity of that ability are unknown.

Section 2: Categorical Perception Revisited

- **Categorical perception** is the pattern in which listeners hear a smoothly changing series of sounds as falling into sharp, separate categories. In the classic demonstration, a set of 14 synthetic speech sounds varies in equal physical steps, and listeners hear each sound as **b**, **d**, or **g**.
    - The 14 stimuli are simplified speech patterns made of two bands of energy called formants, plotted with frequency in cycles per second (CPS, from 0 to 2400) against time in milliseconds, with each sound lasting about 300 milliseconds. The lower formant is the same in all 14 stimuli, rising quickly at the start to a steady level near the bottom of the frequency range.
    - What changes from stimulus to stimulus is the starting portion of the upper formant, called the transition. In stimulus 1, the upper formant starts low and rises steeply to its steady level near 2000 CPS. Across stimuli 2 through 7, that rise becomes smaller and smaller until the transition is nearly flat, and across stimuli 8 through 14, the transition starts higher than the steady level and falls down to it, with the fall getting steeper until stimulus 14 has the sharpest drop.
- The **labelling data** record the percentage of times listeners identified each stimulus as b, d, or g, with stimulus value from 1 to 14 on the horizontal axis and percent identification from 0 to 100 on the vertical axis.
    - Stimuli 1 through 3 are identified as **b** close to 100% of the time. At stimulus 4, b responses drop to about 30%, and by stimulus 5 they reach zero.
    - Over that same short stretch, **d** responses climb from almost zero at stimulus 3 to about 70% at stimulus 4 and to 100% at stimulus 5. Stimuli 5 through 8 are heard as d essentially all of the time, then d responses fall to about 85% at stimulus 9, about 25% at stimulus 10, and close to zero by stimuli 11 and 12.
    - **g** responses are at zero through stimulus 8, rise to about 15% at stimulus 9 and almost 80% at stimulus 10, and stay near 100% from stimulus 11 through 14.
    - The physical change from one stimulus to the next is the same size everywhere along the series, but the labels do not change gradually. Each label holds steady across a wide stretch of stimuli and then flips within one or two steps, which places one category boundary between stimuli 3 and 5 and a second boundary between stimuli 9 and 11.
- **Discrimination** was tested by asking listeners to tell apart two stimuli separated by one, two, or three steps along the series, and these are the **one-step**, **two-step**, and **three-step discrimination** conditions. Each pair is plotted by the value of its first member, called the **"A" stimulus**, with percent correct on the vertical axis, where 50% is the level of guessing.
    - Two curves appear in each discrimination condition. The **obtained** curve is how well listeners actually performed, and the **predicted** curve is how well they would perform if they could only tell two sounds apart when they gave them different labels.
    - In the **two-step** condition, obtained accuracy reaches about 100% for the pair starting at stimulus 3, which is the pair that crosses the b and d boundary, and about 98% for the pair starting at stimulus 8, which crosses the d and g boundary. For pairs starting at stimuli 5 and 6, both of which fall inside the d category, accuracy drops to around 50%, which is chance. The predicted curve follows the same peaks and valleys, although obtained accuracy also stays high,

between about 80% and 90%, for pairs starting at stimuli 10 through 12, where the prediction stays near 50%.
    - In the **one-step** condition, accuracy is lower overall since the two sounds are physically closer together. Obtained accuracy peaks at about 78% to 80% for pairs starting at stimuli 3 and 4 and at about 73% for the pair starting at stimulus 8, and it sits near 50% for most within-category pairs.
    - In the **three-step** condition, obtained accuracy is near 100% for the pair starting at stimulus 2 and again near 100% for pairs starting around stimulus 7, with high values near the second boundary as well. The predicted curve has peaks at the same boundary locations and drops to 50% for within-category pairs starting at stimuli 5 and 6 and at stimuli 10 and 11.
    - Across all three conditions, listeners discriminate pairs that cross a category boundary much better than pairs of the same physical distance that fall inside one category. Discrimination therefore follows the labels, which is the defining feature of categorical perception.
- **Early theories** of categorical perception treated it as both a **language-specific** and a **species-specific** phenomenon. At that time, only humans were known to show categorical perception, and they were known to show it only for speech sounds.
- Categorical perception is a compelling example of how **similarity between stimuli is defined in linguistic, rather than physical, terms**. Two sounds that are physically very different can be heard as the same if they share a category label, while two sounds that are physically very close can be heard as clearly different if they fall on opposite sides of a category boundary.

Section 3: Methods for Testing Infant Speech Perception

- Testing categorical perception in **infants** requires methods that do not depend on the infant saying anything or pressing buttons. One such method is the **conditioned head turn procedure**, used in **Janet Werker's** infant studies laboratory (http://infantstudies.psych.ubc.ca/).
    - In this setup, the infant sits on a parent's lap facing an experimenter, who keeps the infant's attention forward with toys on a small table. To the infant's side is a display box with dark panels, and when one of its compartments lights up, a toy inside becomes visible, which gives the infant a reason to turn toward it. Both the parent and the experimenter wear headphones during testing.
    - The procedure is called a **conditioned** head turn because the infant's head turn toward the display is a learned response, and that response is what researchers use to measure whether the infant noticed something about the sounds.
- A second method is **"high amplitude sucking."** In this method, the infant sucks on a pacifier while the strength and timing of each suck are recorded.
    - The sucking record is a continuous trace of repeated peaks, where each peak is one strong suck. Researchers count how many of these sucking responses the infant makes per minute and track how that rate changes over the course of the experiment.
    - The example of a sucking record comes from **Rochat & Striano (2001)**, published in *Infant Behavior and Development* (IBD), a paper unrelated to the speech perception studies.

Section 4: Categorical Perception in Four-Month-Olds (Eimas et al., 1971)

- **Eimas et al. (1971)**, published in *Science*, used the high amplitude sucking procedure to test whether very young infants perceive speech sounds categorically. The sounds varied in **voice onset time (VOT)**, which is the time between the moment a consonant is released and the moment the vocal folds begin vibrating.

    - The stimuli were synthetic syllables built from three formants, labeled **F-1**, **F-2**, and **F-3**, plotted with frequency in kilocycles from 0 to 3 against time. In a syllable with a VOT of **+10 msec**, all three formants begin almost at once, just 10 milliseconds after the release.
    - In a syllable with a VOT of **+100 msec**, the upper two formants begin at the release, but the regular vibration and the first formant, F-1, do not start until 100 milliseconds later. During that delay, F-2 and F-3 carry a noisier, weaker form of energy before the full voiced portion of the syllable begins.
- The main experiment had **three variations in VOT differences** at each of **two age levels**, and in each condition the infant heard one stimulus for a period of time and then heard a second stimulus after a shift.
    - In condition **20D**, the two stimuli differed in VOT by **20 msec** and came from **different adult phonemic categories**, meaning adults would hear them as two different consonants. The two stimuli in this condition had VOT values of **+20** and **+40 msec**.
    - In condition **20S**, the VOT difference was again **20 msec**, but the two stimuli came from the **same phonemic category**, meaning adults would hear them as the same consonant. The stimuli in this condition had VOT values of **−20 and 0 msec** or **+60 and +80 msec**.
    - Condition **0** was a **control condition**. In the control group, there was no actual change in the stimulus, and the time at which a shift would have occurred is marked for comparison.
- The key data come from the **4-month-old infants**, plotted as the **mean number of sucking responses** per minute, on a vertical axis from 15 to 75, as a function of time and condition. The letter **B** marks the **baseline rate**, which is the sucking rate before any sounds began, and time is measured relative to the moment of stimulus shift, covering the 5 minutes before the shift and the 4 minutes after it, with a dashed line marking the shift itself.
    - In every condition, sucking starts at a baseline of roughly 24 to 26 responses per minute, then jumps well above baseline once the first stimulus begins, and then declines over the minutes just before the shift. In condition **20D**, sucking rises to about 52 to 55 responses per minute and then falls to about 34 by the last minute before the shift. In condition **20S**, it rises as high as about 66 and falls to about 35, and in the **control** condition it rises to about 58 and falls to about 30.
    - After the shift, the three conditions separate. In **20D**, sucking jumps back up to about 51 responses in the first minute after the shift and stays near 47 or 48 for the remaining minutes, which means the infants responded to the new sound as something new.
    - In **20S**, sucking moves only slightly at the shift, to about 38 responses, and then keeps falling to about 28. In the **control** condition, sucking keeps dropping steadily, from about 27 down to about 16 by the fourth minute.
    - The 20D and 20S conditions involve exactly the same 20-millisecond physical difference in VOT. The infants' sucking recovered only when that difference crossed an adult category boundary, so 4-month-olds were discriminating speech sounds categorically, in the same way adults do.

Section 5: Why Infant Categorical Perception Seemed Like Evidence for Innateness

- Categorical perception was considered to be the **first step in turning speech into a symbolic format** that could be used by later linguistic processes. Once a continuously varying sound is converted into a category such as b or d, later stages of language processing can work with those discrete symbols rather than with raw acoustic detail.
- Four-month-olds are very young, and their general development makes this point clear. Some **gross motor milestones** for that age include **manipulating objects with two**

hands and **holding their head and chest up while prone**, which means lying on the stomach.
    - Their **color vision** and **general visual acuity** are still "in progress," meaning these abilities are not yet fully developed.
    - In terms of their own vocal output, they are just starting to produce **vowel-like sounds** instead of **creaky whines**.
- Finding categorical perception at 4 months meant that an apparently **language-specific ability** was emerging very, very early in development. Infants at this age have had very little time to learn from experience, and they cannot yet produce speech themselves, so the early appearance of categorical perception seemed like **evidence for innateness**.

Section 6: Perceptual Narrowing for Speech Categories

- At around **six months**, English-learning infants show evidence of discriminating speech contrasts they have never heard before. Over the course of development, the ability to perceive these contrasts **declines**, and this decline is called **"perceptual narrowing"** for speech categories.
- **Werker & Tees (1984a)**, in the paper "Cross-language speech perception: Evidence for perceptual reorganization during the first year of life," published in *Infant Behavior and Development*, volume 7, pages 49–63, tested the effects of age and experience on infants' ability to discriminate **two non-English contrasts**.
    - The first contrast was the **Hindi Ta/ta** contrast, which is a distinction between two different t-like sounds that Hindi treats as separate. The second was the **Salish ki/qi** contrast, which is a distinction between two different k-like sounds in Salish.
    - The measure was the **percentage of infants at each age reaching a discrimination criterion**, meaning the proportion of infants whose responses met the standard for showing that they could tell the two sounds apart.
- The results changed sharply with age for English-learning infants.
    - Among **English infants aged 6–8 months**, about 95% reached criterion on the Hindi contrast and about 80% on the Salish contrast. At this age, most English-learning infants could discriminate both contrasts even though neither one occurs in English.
    - Among **English infants aged 8–10 months**, the numbers fell to about 67% for Hindi and about 52% for Salish.
    - Among **English infants aged 10–12 months**, only about 20% reached criterion on the Hindi contrast and about 10% on the Salish contrast.
    - **Hindi and/or Salish infants**, who were learning a language that uses these contrasts, reached criterion at close to 100% on both. Infants with experience of the contrasts kept the ability, while English-learning infants lost most of it by the end of the first year of life.

Section 7: The Nativist Narrative of Speech Perception

- The findings on infant categorical perception and perceptual narrowing fit together into a very compelling narrative. In this narrative, **speech perception depends on categorical perception abilities** that are a **special property of the language system**.
    - These categorical perception abilities are present **very early in development**, as the 4-month-olds in Eimas et al. (1971) indicated.
- In fact, according to this account, children start off as **"citizens of the world,"** able to hear **all contrasts**. This means they have **innate, language-general knowledge about speech**, which is knowledge of the full set of sound distinctions that any human language might use.

    - Children then **rapidly tune their perceptual systems** to the **specific language in their environment**, which is what the decline in English-learning infants' discrimination of Hindi and Salish contrasts reflects.
- Under this narrative, **the role of learning** is to **select** among the many possible **innately specified phonetic contrasts**. Learning narrows the inventory down to just those contrasts that are **meaningful in the child's native language**.
    - In this view, learning does not create new categories. All of the categories are already provided, and experience only determines which ones are kept and which ones are lost.

Section 8: Problems for the Nativist View

- The nativist narrative faces several problems on the side of **categorical perception** itself.
    - Categorical perception is **not limited to language**. It also appears for sounds that are not speech, which goes against the early claim that it is a language-specific phenomenon.
    - Categorical perception is **not limited to humans**. Non-human animals can also show it, which goes against the early claim that it is species-specific.
    - Categorical perception is **task-dependent**. Whether listeners appear to perceive sounds categorically depends on the task used to measure their perception.
- The narrative also faces a problem on the side of **infant speech perception**. Infant speech perception is **not actually "universal,"** meaning infants do not start out able to hear every possible contrast equally well, which goes against the idea that children begin as citizens of the world with access to all contrasts.

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Section 9: Categorical Perception of Non-Speech Sounds (Cutting & Rosner, 1974)

- **Cutting & Rosner (1974)**, published in *Perception & Psychophysics*, tested whether categorical perception appears for sounds that are not speech. The sounds varied in **rise time**, which is how long it takes a sound to grow from silence to its full loudness.
    - Their Experiments I and II used three types of sounds: **sawtooth waves**, **speech syllables**, and **sine waves**. Each type was made with different rise times, and sample versions at a **10 msec rise time** and a **70 msec rise time** are compared, with each sound lasting roughly 400 msec.
    - At a 10 msec rise time, each sound reaches full strength almost instantly, so its waveform begins abruptly and then tapers off. At a 70 msec rise time, the same sound swells more gradually before reaching full strength.
    - For the sawtooth waves, the rise time determines whether the sound resembles a **plucked** string or a **bowed** string. A short rise time sounds like a pluck, and a long rise time sounds like a bow.
- Listeners labeled sawtooth waves with rise times from 0 to 80 msec, in steps of 10 msec, as either **PLUCK** or **BOW**, with percent correct identification on the left vertical axis.
    - **PLUCK responses** are near 98% at rise times of 0 and 10 msec, fall to about 88% at 20 msec and about 80% at 30 msec, then drop to about 50% at 40 msec. They continue falling to about 25% at 50 msec and reach only a few percent by 70 and 80 msec.
    - **BOW responses** follow the mirror image of that pattern. They are near zero at 0 and 10 msec, rise to about 10% at 20 msec and about 18% at 30 msec, reach about 50% at 40 msec, and then climb to about 72% at 50 msec and above 90% by 80 msec.
    - The two label curves cross at a rise time of about **40 msec**, which places the category boundary between pluck and bow at that point.

- Listeners also tried to **discriminate** pairs of sawtooth waves whose rise times differed by 20 msec, with percent correct discrimination on the right vertical axis. The pairs were 0–20, 10–30, 20–40, 30–50, 40–60, 50–70, and 60–80 msec.
    - Discrimination is around 58% for the 0–20 and 10–30 pairs, both of which fall within the pluck category. It rises to about 72% for the 20–40 pair and peaks at about 76% for the 30–50 pair, which are the pairs that span the boundary near 40 msec.
    - For the 40–60, 50–70, and 60–80 pairs, which fall within the bow category, discrimination drops back to about 55%.
    - Every pair has the same 20 msec physical difference, yet listeners discriminate best right at the pluck and bow boundary and poorly within each category. Sharp labeling plus a discrimination peak at the boundary is the same categorical pattern found for speech, which means categorical perception is **not limited to language**.

Section 10: Categorical Perception of Non-Speech Sounds by Infants (Jusczyk et al., 1977)

- **Jusczyk et al. (1977)**, published in *Perception & Psychophysics*, used the high amplitude sucking procedure to test whether **infants** also perceive non-speech sounds categorically. The study compared three experimental sessions: **No Change**, **Within Categories**, and **Between Categories**.
    - In the No Change session, the infant kept hearing the same sound throughout. In the Within Categories session, the sound changed at the shift to a different sound from the same category, and in the Between Categories session, the sound changed to one from a different category.
- The results are plotted as **mean sucking rates**, on a vertical axis from 3 to 30 responses per minute, as a function of time and session. Time is measured relative to the moment of the stimulus shift, which is marked by a vertical dashed line, with the 3 minutes before the shift and the 4 minutes after it, and the letter **B** marks the baseline rate of sucking.
    - In all three sessions, sucking begins at a baseline of roughly 13 to 15 responses per minute, rises to about 20 or 21 in the third minute before the shift, and then falls to about 11.5 to 12.5 by the last minute before the shift.
    - In the **No Change** session, sucking stays flat at about 12 responses per minute for all 4 minutes after the shift time. In the **Within Categories** session, sucking also stays flat, at about 11 to 12.5 responses per minute, even though the sound actually changed.
    - In the **Between Categories** session, sucking recovers to about 17.5 responses in the first minute after the shift and about 16 in the second minute, before returning to about 12.5.
    - Infants responded to a change in the sound only when the change crossed a category boundary, and they treated a within-category change as if nothing had changed. Infants therefore show categorical perception for non-speech sounds too, so the early appearance of categorical perception in infancy is not specific to speech.

Section 11: Categorical Perception of Speech Sounds by Chinchillas (Ohlemiller et al., 1999)

- **Ohlemiller et al. (1999)**, published in *Behavioral Brain Research*, tested categorical perception of **speech** sounds in **chinchillas**, building on earlier work by **Kuhl & Miller (1975)**. The chinchillas were trained using **operant conditioning**, the same reward-based learning that behaviorists studied.
    - In training, either the **0-lag** stimulus or the **80 ms-lag** stimulus was made an S^D for food. An S^D, or discriminative stimulus, is a signal that tells the animal a response will now be rewarded. The 0-lag and 80 ms-lag stimuli are the two

ends of a voice onset time continuum, with 0 and 80 milliseconds between the release of the consonant and the start of voicing.
    - Once trained on these endpoints, the animals were tested on the stimuli in between, which reveals how they divide up the continuum.
- The stimuli formed **/ga/–/ka/ continua**, and both chinchillas and humans were tested on them. The measure is the **proportion of stimuli labeled /ga/**, from 0 to 1, plotted against **VOT** from 0 to 120 ms.
    - There were three versions of the continuum. One ran from 0 to 80 ms, another ran from 0 to 120 ms, and a third used **enhanced burst stimuli**, in which the intensity of the **frication burst** and **aspiration noise** at the start of the consonant was increased. For each version, the individual responses are given as raw data points, and a smooth fitted curve is drawn through them.
    - Four listeners are compared: two chinchillas, **Animal C1** and **Animal C3**, and two humans, **Subject S3** and **Subject S4**.
- All four listeners produce sharp, step-like labeling curves, with short-VOT stimuli labeled /ga/ and long-VOT stimuli labeled /ka/.
    - **Animal C1** labels stimuli up to about 30 ms as /ga/ around 90% of the time, then drops steeply, crossing 0.5 near 45 ms and leveling off around 0.15 to 0.2 at longer VOTs. The three versions of the continuum produce nearly identical curves for this animal.
    - **Animal C3** labels short VOTs as /ga/ more than 90% of the time and then declines more gradually than C1. For this animal, the enhanced burst curve crosses 0.5 at a shorter VOT than the 0–80 ms curve, and the 0–120 ms curve declines the most gradually, still above 0.1 near 120 ms.
    - **Subject S3** labels stimuli up to about 50 ms as /ga/ nearly 100% of the time and then drops steeply to near zero by about 80 ms, with the boundary in the low 60s of milliseconds. For this human listener, the enhanced burst curve is steeper than the standard curve, while the 0–120 ms curve falls a little later.
    - **Subject S4** labels stimuli up to about 30 ms as /ga/ nearly 100% of the time and drops almost straight down to near zero by about 45 to 50 ms, with the three versions nearly overlapping.
    - Increasing the intensity of the burst and aspiration noise **shifted the category boundary to a shorter VOT and/or increased the slope of the curve** for one animal, C3, and one human listener, S3.
- The chinchilla curves are just as categorical as the human curves, with boundaries in a similar range of VOT values, and the burst manipulation affected one chinchilla and one human in the same way. Chinchillas are not humans and have no language, so categorical perception of speech sounds is **not limited to humans**.

Section 12: Some Contrasts Are Less Salient Than Others for Infants (Narayan et al., 2010)

- **Narayan et al. (2010)** tested whether infants can perceive all speech contrasts equally well from the start. They measured **looking time** in seconds on **same** trials, where the infant hears one syllable repeated, and on **change** trials, where the syllable switches. Longer looking on change trials than on same trials means the infant noticed the switch.
- The first test used **English-hearing infants at 4–5 months**, who were tested on two contrasts. The **[ma]–[na]** contrast is native to English, and the **[na]–[ŋa]** contrast is non-native, with error bars representing standard error.
    - For **[ma]–[na]**, looking time rises from about 3.4 seconds on same trials to about 5.4 seconds on change trials. The infants clearly detected this contrast.
    - For **[na]–[ŋa]**, looking time is about 4.5 seconds on same trials and about 4.2 seconds on change trials, so there is no increase when the sound changes. The infants did not detect this contrast.

    - The [na]–[ŋa] pair is a **place contrast for nasals in onset**, meaning the two sounds are both nasal consonants at the start of a syllable, and they differ in where in the mouth the sound is made. This contrast, written /**na**/–/**ŋa**/, is not readily perceived by four-month-olds, **independent of language experience**, while /**ma**/–/**na**/ is relatively salient, meaning it is easy for infants to notice.
- The second test used **Filipino-learning infants at 6–8 months and 10–12 months**, tested on **[na]–[ŋa]**, which is a native contrast in their language environment.
    - The **6–8-month-olds** look for about 3.8 seconds on both same and change trials, so even these infants, who hear the contrast in their own language, do not yet detect it.
    - The **10–12-month-olds** look for about 3.3 seconds on same trials and about 4.8 seconds on change trials. By this age, the Filipino-learning infants detect the contrast.
    - Between 6 and 10 months, exposure to this contrast in a **Filipino language environment** increases its salience.
- These results mean that not all speech contrasts are available innately. Infants are not born hearing every contrast, so infant speech perception is not actually universal, and **exposure to contrasts in the environment is necessary for acquisition** in some cases. In these cases, experience does more than select from a built-in set of contrasts, since the ability to hear the contrast appears only after the infant has been exposed to it.

Section 13: Emergentist Approaches and Associative Learning

- The alternative to the nativist approach is the **emergentist approach**. Emergentist approaches assume that **associative learning mechanisms** are sensitive to **statistical regularities in the environment**, which are the patterns in how often things occur and how often they occur together.
- Associative learning at the level of neurons involves three cells in a classical conditioning setup: a **Bell** neuron, a **Food detector** neuron, and a **Salivation** neuron.
    - The food detector neuron is connected to the salivation neuron, so detecting food already triggers salivation, and the signal flows in the direction from the input neurons toward the salivation neuron.
    - The bell neuron also sends a connection to the salivation neuron, and this connection is labeled the **synapse to be strengthened**. When the bell neuron and the salivation neuron are active together, which happens when the bell rings at the same time food is detected, this synapse grows stronger.
    - Once the synapse is strong enough, the bell alone can activate the salivation neuron. Learning in this system consists of changing connection strengths based on which events reliably occur together.
- The challenge for the emergentist approach is how this kind of associative mechanism can extend beyond **conditioning**, where one stimulus becomes linked to one response, to support learning for **pattern recognition**, such as recognizing the sound categories of a language.

Section 14: Vowel Distributions and Statistical Learning

- When **Kuhl** and others talk about learning the **statistics of distributions in the environment**, they mean measurements like those of **American English vowels** plotted in a **"vowel space."** The vowel space is determined by the **first formant** (F1) and the **second formant** (F2), which are two of the main frequency bands in a vowel.
    - The horizontal axis gives the first formant from 300 to 1,200 Hz, and the vertical axis gives the second formant from 600 to about 3,400 Hz. Each point is one spoken vowel, marked with the symbol for the vowel the speaker intended, and each vowel category has an ellipse drawn around its cloud of points.

    - The vowel marked **i** sits in the upper left, with a low first formant of roughly 300 to 500 Hz and a very high second formant reaching above 3,000 Hz. The vowel marked **u** sits in the lower left, with a similarly low first formant but a much lower second formant of roughly 850 to 1,600 Hz.
    - The vowel marked **a** stretches furthest to the right, with first formant values reaching up to about 1,200 Hz. Other categories, including those marked **ae**, **I**, **3**, **U**, ^, and **c**, fill the space between these extremes.
- Each vowel category is not a single point but a **distribution**, which is a spread-out cloud of many different productions. These clouds are wide, and many of them **overlap**, so a single combination of F1 and F2 values can fall inside the ellipses of two or three different vowels at once. This overlap is the reason speech categorization is difficult.
- According to the statistical learning idea, we might learn the vowels of our language by **extracting the statistics of these distributions** from the speech we hear.
    - Under this idea, a **"prototypical" /a/** is just the **mean** of the distribution of all the /a/ sounds we hear, or possibly the **mode**, which is the most frequent value. The best example of a category is not stored innately but is computed from the input.

Section 15: Distributional Learning in Infants (Maye et al., 2002)

- **Maye et al. (2002)**, published in *Cognition* (volume 82, pages B101–B111), tested whether infants actually learn categories from the statistics of the sounds they hear. Infants **listened passively** to one of two distributions of **voice onset time** along a continuum from **[da]** to **[ta]**.
    - The continuum had 8 tokens, with **Token 1** being the endpoint **[da]** stimulus and **Token 8** being the endpoint **[ta]** stimulus. During the **familiarization** phase, each token was played a certain number of times, and that number is the **familiarization frequency**.
- The two groups heard different distributions of these tokens.
    - The **bimodal** group heard a distribution with two peaks, which provided evidence for **two different categories**. Tokens 1 and 8 were each played 4 times, tokens 2 and 7 were each played 16 times, tokens 3 and 6 were each played 8 times, and tokens 4 and 5 were each played 4 times, so most of the input clustered near the two ends of the continuum.
    - The **unimodal** group heard a distribution with one peak, which provided evidence for a **single category**. Tokens 1, 2, 7, and 8 were each played 4 times, tokens 3 and 6 were each played 8 times, and tokens 4 and 5 were each played 16 times, so most of the input clustered in the middle.
    - Tokens 3 and 6 were each played 8 times in both groups. The two groups heard these particular tokens equally often and differed only in the overall shape of the distribution around them.
- At test, infants listened to **stimuli 3 and 6** in either **alternating trials**, where the two stimuli switched back and forth, or **non-alternating trials**, where one stimulus was repeated. The measure was mean looking time in seconds, with standard error in parentheses.
    - **6-month-old unimodal** infants looked 4.85 s (0.47) on alternating trials and 4.53 s (0.51) on non-alternating trials.
    - **8-month-old unimodal** infants looked 4.98 s (0.63) on alternating trials and 5.20 s (0.56) on non-alternating trials.
    - **6-month-old bimodal** infants looked 5.66 s (0.44) on alternating trials and 6.41 s (0.32) on non-alternating trials.
    - **8-month-old bimodal** infants looked 5.45 s (0.52) on alternating trials and 6.15 s (0.56) on non-alternating trials.

- Only infants pre-exposed to the **bimodal distribution** were sensitive to the difference between the alternating and non-alternating conditions, with looking times differing by about 0.7 seconds at both ages. Unimodal infants had small differences in opposite directions at the two ages.
    - The unimodal infants behaved as if they had learned to treat stimuli 3 and 6 as **members of the same category**, while the bimodal infants behaved as if 3 and 6 belonged to different categories. A few minutes of passive listening to a particular distribution changed how infants at 6 and 8 months grouped the same pair of sounds.

Section 16: Social and Interactive Context for Learning

- Experimental evidence indicates that **social interaction** is important for speech learning. The same exposure to a **Mandarin** speaker is more effective for learning when it happens **in person** and is interactive than when it is viewed as a **video**.
- In the first experiment, **nine-month-old American infants** were exposed to **Mandarin Chinese** in **twelve 25-minute sessions**.
    - One group received **live exposure**, with a Mandarin speaker playing and reading with the infants in person. Other groups received **auditory or audiovisual exposure**, meaning they heard the speaker through audio alone or watched the speaker on a television screen.
    - After exposure, the **Mandarin exposure groups** and **English control groups** were tested on a **Mandarin phonetic contrast** using the **head-turn procedure**. The English control infants received the same kind of sessions in English rather than Mandarin.
    - In the live condition, infants exposed to Mandarin reached about 66% correct, while English control infants reached about 56%, with chance at 50%. Infants in the **TV** and **audio** conditions each reached about 56%, the same as the English controls.
    - Phonetic learning occurred in the **live-exposure group**, but there was **no learning** in the TV or audio-only groups. Hearing the same Mandarin sounds was not enough on its own, and learning occurred only when the speaker was physically present.
- **Social interactions contingent on the production of different sounds** seem to function much like a **reinforcing stimulus**. Contingent means that the social response depends on, and follows, the infant's own behavior.
    - In the second experiment, **eight-month-old infants** received either **contingent** or **non-contingent social feedback** from their mothers in response to their vocalizations.
    - In the **contingent condition (CC)**, Visit 1 consisted of a 30-minute **familiarization** period. Visit 2 consisted of a 10-minute **baseline**, then 10 minutes of **contingent social response**, in which the mother responded right after the infant vocalized, and then a 10-minute **extinction** period, in which the social responses stopped.
    - In the **yoked-control condition (YC)**, the schedule was the same, except that the middle 10 minutes involved **non-contingent social response**. These infants received social responses that were not timed to their own vocalizations.
- The measure was the **proportion of syllables to vocalizations**, meaning how many of the infant's vocalizations contained syllables, which are more complex, speech-like sounds.
    - At **baseline**, the proportion is about 0.08 for the contingent group and about 0.04 for the non-contingent group.
    - During the **social response** period, the contingent group rises to about 0.32, while the non-contingent group stays around 0.05.

    - During **extinction**, the contingent group falls to about 0.20, still well above its baseline, and the non-contingent group is about 0.09.
    - **Contingent social feedback increased the quantity and complexity of infants' vocalizations**. The social responses acted like reinforcement, since the infants changed their vocal behavior only when the responses followed their own sounds.
    - The phonetic learning results and the social feedback results both come from studies published in 2003 by the National Academy of Sciences USA.
- There are two ideas about why social interaction matters for learning.
    - One idea is that humans find social interaction with other humans **innately reinforcing**. In this view, social contact works like a built-in reward that strengthens whatever behavior it follows.
    - Another idea is that humans are **innately good at using social cues** to learn from other people. In this view, social interaction helps by guiding the learner toward the relevant information, such as what the speaker is attending to.

Section 17: Statistical Learning with Constraints

- The emergentist view does not claim that learners absorb every statistical pattern in the input equally. Instead, it involves **statistical learning with constraints**, and "constraints" refers to a lot of things.
- The first kind of constraint concerns **what stimulus properties can be contrastive**. Some properties of speech vary a great deal from one utterance to the next but never distinguish one word or sound category from another, such as **who the speaker is** and **speech rate**. A learner has to track the statistics of the properties that can signal a contrast while not forming categories around properties like these.
- The second kind of constraint is based on **earlier learning**. What a learner has already learned shapes what the learner can learn next, and constraints of this kind are connected to **sensitive periods** in development.
- The third kind of constraint is the **social and/or interactive context for learning**. Statistical regularities in speech are learned more effectively when they come through live, interactive, and contingent social exchanges, as in the live Mandarin exposure and contingent maternal feedback experiments, than when the same input is presented without that social context.